

3.2 The Derivative as a Function

In the last section we defined the derivative of $y = f(x)$ at the point $x = x_0$ to be the limit

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

We now investigate the derivative as a function derived from f by considering the limit at each point x in the domain of f .

① The Derivative of $f(x)$ with Respect to x

DEFINITION The **derivative** of the function $f(x)$ with respect to the variable x is the function f' whose value at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists.

- ◆ If f' exists at a particular x , we say that f is differentiable (has a derivative) at x . If f' exists at every point in the domain of f , we call f differentiable.

The process of calculating a derivative is called differentiation. To emphasize the idea that differentiation is an operation performed on a function $y = f(x)$, we use the notation

$$\frac{d}{dx} f(x)$$

as another way to denote the derivative $f'(x)$.

Example 1 Differentiate $f(x) = \frac{x}{x-1}$.

Example 2

- (a) Find the derivative of $f(x) = \sqrt{x}$ for $x > 0$.

(b) Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

② Some Common Notations for the Derivative of $y = f(x)$

Some common alternative notations for the derivative are

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = D(f)(x) = D_x f(x).$$

The symbols $\frac{d}{dx}$ and D indicate the operation of differentiation. The “prime” notations y' and f' come from notations that Newton used for derivatives.

To indicate the value of a derivative at a specified number $x = a$, we use the notation

$$f'(a) = y'(a) = \left. \frac{dy}{dx} \right|_{x=a} = \left. \frac{df}{dx} \right|_{x=a} = \left. \frac{d}{dx} f(x) \right|_{x=a}.$$

③ One-Sided Derivatives

A function $y = f(x)$ is **differentiable on an open interval** (finite or infinite) if it has a derivative at each point of the interval. It is **differentiable on a closed interval** $[a, b]$ if it is differentiable on the interior (a, b) and if the limits

$$\begin{aligned} f'(a^+) &= \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} && \text{Right-hand derivative at } a \\ f'(b^-) &= \lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h} && \text{Left-hand derivative at } b \end{aligned}$$

exist at the endpoints.

For general point a , we have

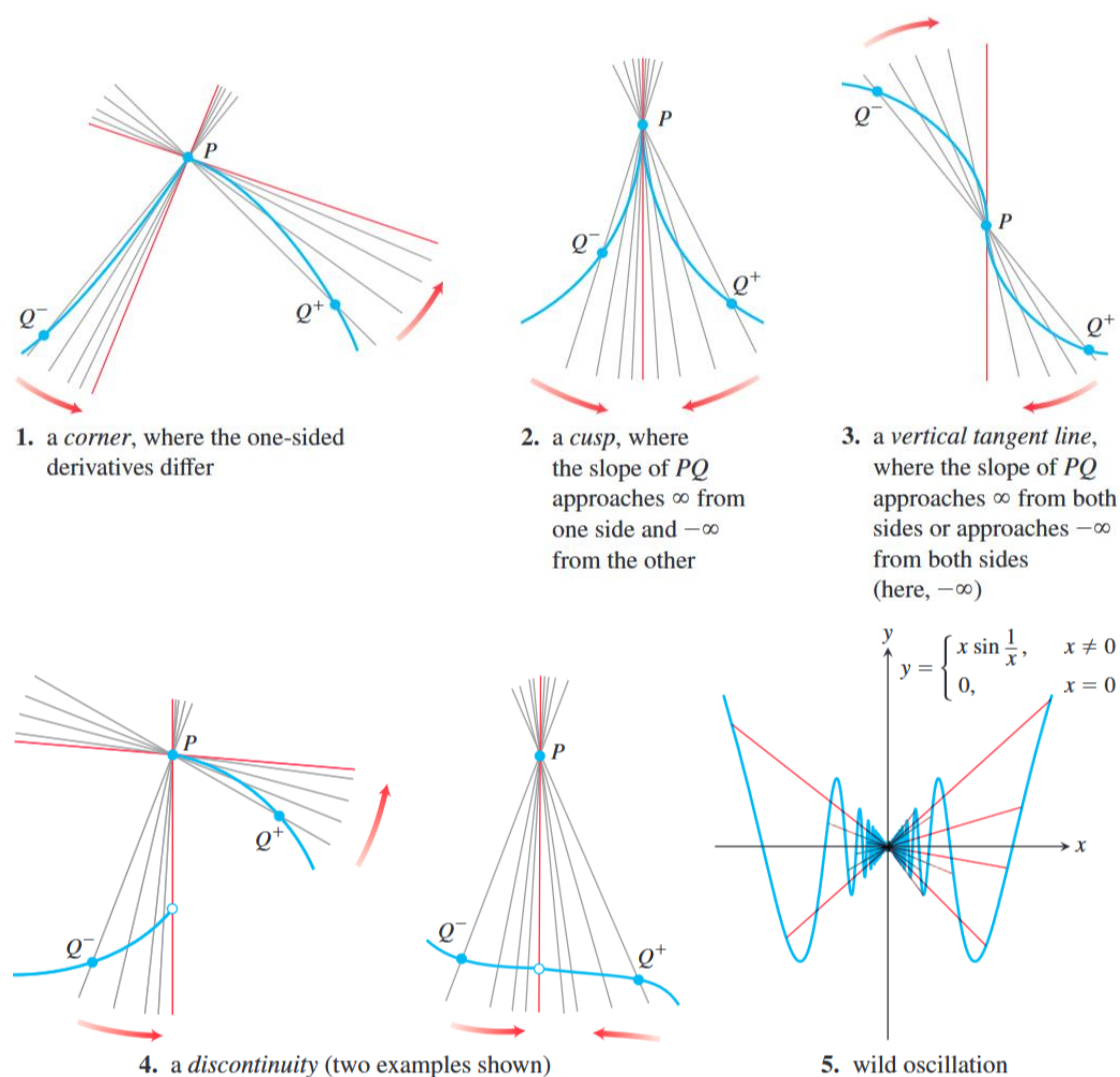
$$f'(a) = L \Leftrightarrow f'(a^+) = f'(a^-) = L.$$

Example 3 Show that the function $f(x) = |x|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$ but has no derivative at $x = 0$.

Example 4 Let $f(x) = \sqrt{x}$. Show that the function $f(x)$ has no derivative at $x = 0$.

④ When Does a Function Not Have a Derivative at a Point?

A function has a derivative at a point x_0 if the slopes of the secant lines through $P(x_0, f(x_0))$ and a nearby point Q on the graph approach a finite limit as Q approaches P . Thus differentiability is a “smoothness” condition on the graph of f . A function can fail to have a derivative at a point for many reasons, including the existence of points where the graph has



⑤ Differentiable Functions Are Continuous

THEOREM 1 If f has a derivative at $x = c$, then f is continuous at $x = c$.

Note The converse of Theorem 1 is false. See example 3 and 4.